Project scope and current status

- Target problem: health and economic impact of introducing PCVs for a single Vietnamese birth cohort over the first 5 years of life.
- Two aligned epidemiological implementations already exist: a stochastic monthly microsimulation and a deterministic cohort ODE expectation analogue.
- Current extension focuses on three tasks: exact analytic characterization of the ODE, structured interpretation of model assumptions, and deterministic plus probabilistic sensitivity analysis.
- A closed-form monthly propagator has now been implemented and numerically validated against the original RK45 solution.

Model structure

State vector:

$$Y(t) = \left(A_v, A_u, C_{men}, C_{psev}, C_{pnon}, C_{sep}, C_{aom}, C_{death}, S_{cog}, S_{hear}, S_{epi}, S_{hemi}, S_{impl}, S_{sped}\right)^\top.$$

- A_v, A_u: alive vaccinated and unvaccinated subcohorts.
- $C_\cdot$: cumulative disease counts and cumulative disease deaths.
- $S_\cdot$: living sequela stocks after meningitis survival.
- Time is continuous in months, but all epidemiological coefficients are monthwise constant because the implementation uses $m = \lfloor t \rfloor$.
- The monthwise-constant structure is the key reason that an exact piecewise analytic solution exists.

Monthly coefficient construction

Within month m, the code builds event probabilities from annual under-five incidence using

$$x_d(m) = \frac{I_d \cdot 5 \cdot w_d(m)}{100000}, \qquad p_d(m) = 1 - e^{-x_d(m)}.$$

These probabilities are converted back to continuous-time hazards,

$$\lambda_d^{(m)} = -\ln(1 - p_d(m)).$$

- Background mortality is represented by a constant monthly hazard derived from the under-five mortality rate.
- Vaccine effect is introduced through annual IRR trajectories and then mapped to vaccinated monthly probabilities.
- Because $w_d(m)$ and the annual vaccine multiplier are fixed within each month, the resulting ODE is a piecewise constant-hazard system.

Exact piecewise analytic solution

For $t = m + \tau$ with $\tau \in [0, 1)$ and month-specific all-cause hazards $\mu_v^{(m)}, \mu_u^{(m)}$,

$$A_v(m + \tau) = A_{v,m} e^{-\mu_v^{(m)} \tau}, \qquad A_u(m + \tau) = A_{u,m} e^{-\mu_u^{(m)} \tau}.$$

For any cumulative endpoint with source coefficients α_v, α_u,

$$C(m + \tau) = C_m + \alpha_v A_{v,m} \phi(\mu_v^{(m)}, \tau) + \alpha_u A_{u,m} \phi(\mu_u^{(m)}, \tau),$$

where

$$\phi(\mu, \tau) = \begin{cases} \dfrac{1 - e^{-\mu\tau}}{\mu}, & \mu > 0, \\ \tau, & \mu = 0. \end{cases}$$

For sequela stocks,

$$\frac{dS}{dt} = \mathrm{src}(t) - \mu_{avg}(t) S(t),$$

which admits an integrating-factor solution because $\mu_{avg}(t)$ is induced by the same exponential alive trajectories.

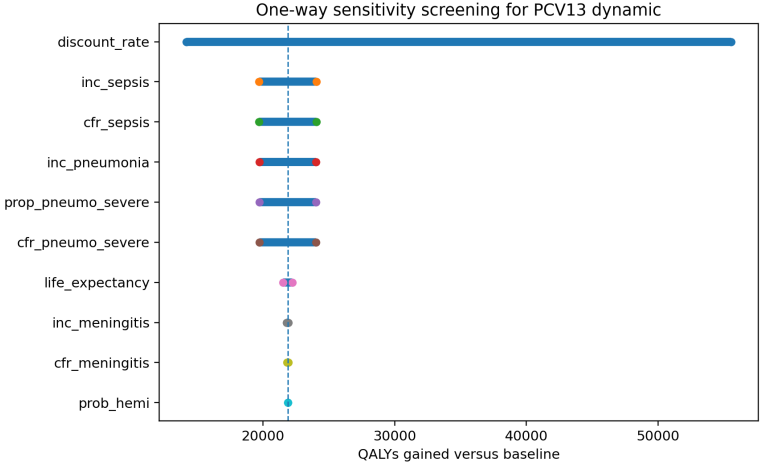
Organizing assumptions into three tiers

1. **Accounting assumptions**: horizon definition, age-weight normalization, reporting boundaries. These do not change the analytic class of the ODE.
2. **Parameter assumptions**: incidence, CFR, utilities, costs, discount rate, life expectancy, and sequela probabilities. These perturb coefficients inside the same closed-form recursion.
3. **Structural assumptions**: vaccine-effect mapping, absence of herd effect, and other rules that determine how coefficients are generated. These can change model behavior at a deeper level.

Evidence already generated:

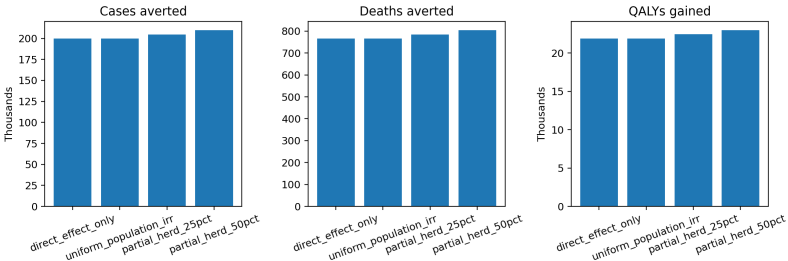
- Age weights sum to 1 for every disease block.
- Prefix invariance is exact for 12, 24, 36, 48, and 60 months.
- Ordinary parameter effects can be ranked by local elasticities.
- Minimal structural alternatives already show meaningful shifts in results.

Parameter sensitivity: which inputs matter most?



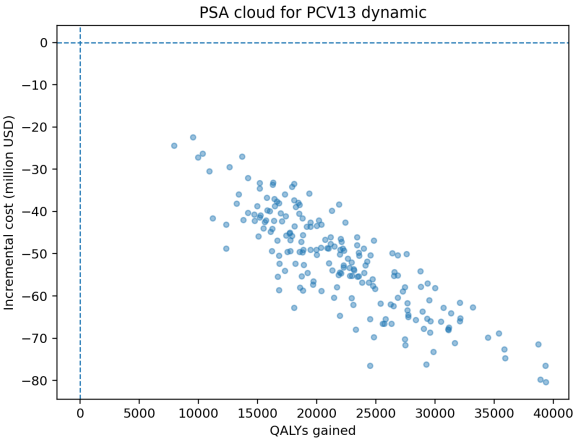
- The largest QALY spreads come from pneumonia and sepsis incidence and CFR terms.
- Discounting changes economic magnitude without changing epidemiological counts.
- Coverage is almost flat under the current mapping because the IRR-to-VE conversion partially offsets the direct effect of changing coverage.

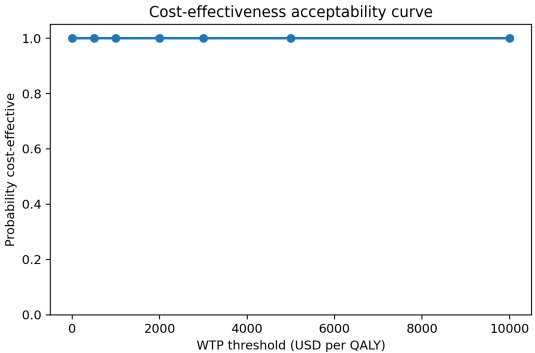
Structural screening



- A simple uniform-population-IRR alternative barely moves the results.
- Partial herd-effect screens increase cases averted, deaths averted, and QALYs gained.
- The no-herd-effect assumption is therefore a meaningful structural choice rather than a harmless simplification.

Probabilistic sensitivity analysis





- Under the current specification, all 200 samples remain in the dominant region for PCV13 dynamic: lower cost and greater health benefit than baseline.
- The CEAC stays at probability 1 over the tested WTP grid.

What the current evidence already proves

1. The monthwise exact ODE implementation is numerically consistent with the original RK45 model.
2. Many assumptions can be separated into accounting, parameter, and structural layers rather than treated as one undifferentiated list.
3. Ordinary parameter uncertainty does not overturn the central cost-saving conclusion in the current ranges.
4. The most promising innovation point is structural, especially vaccine-effect mapping and herd-effect handling, because those choices can move the results without destroying analytic tractability.